# Emmanuel Mombo-Dingba


**Computer Engineer | AI & Machine Learning Engineer**

**Currently AI Engineer @ HADRON S.A.**

Yaounde, Cameroon • +237 653 867 570 • mombodingba.emmanuel@gmail.com
github.com/MOMBO-DINGBA • linkedin.com/in/emmanuel-mombo-dingba • mombo-dingba.github.io


SCAN PORTFOLIO
mombo-dingba.github.io

## PROFILE

Computer Engineer (ENSPY) and AI Engineer at HADRON S.A. with proven expertise in Artificial Intelligence, Machine Learning, Multi-Agent Systems, and enterprise Large Language Models (LLMs). Experienced in designing and integrating AI-driven solutions into production systems (ERP, microservices, real-time event-driven pipelines). Solid mathematical foundation (Linear Algebra, Calculus, Probability, Statistics, Optimization) combined with advanced proficiency in Python, FastAPI, Docker, and modern ML/DL frameworks.

## EDUCATION

**National Advanced School of Engineering (ENSPY)** — Yaounde, Cameroon
*Engineering Degree in Computer Engineering* — **2021 – 2026**

**University of Ngaoundere** — Ngaoundere, Cameroon
*Undergraduate Studies in Pure Mathematics* — **2019 – 2021**

## TECHNICAL SKILLS

**Programming:** Python, SQL, JavaScript, TypeScript, Java, Bash
**AI & Machine Learning:** Machine Learning, Deep Learning, NLP, Large Language Models (LLMs), RAG, Multi-Agent Systems
**Deep Learning:** Neural Networks, CNNs, RNNs, LSTMs, Transformers, Attention Mechanisms
**Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, LangGraph, Pandas, NumPy, OpenCV
**Data & Visualization:** PostgreSQL, MySQL, Redis, Redis Streams, Kafka, Matplotlib, Seaborn
**Software & Deployment:** Git, GitHub, GitLab, Docker, Linux, REST APIs, Microservices, Celery, Jupyter, Google Colab

## PROFESSIONAL EXPERIENCE

**AI Engineer** **[CURRENT POSITION]** — **Feb 2026 – Present**
***HADRON S.A.*** — Yaounde, Cameroon

- Contributed to the design and implementation of an intelligent logistics platform integrated with an enterprise ERP system.
- Designed and integrated AI-driven components to support the automation and optimization of logistics workflows.
- Worked on a Multi-Agent System (MAS) architecture coordinating intelligent services and automating business processes.
- Integrated a conversational AI assistant based on Large Language Models (LLMs) and NLP for natural-language interaction and planning assistance.
- Contributed to the software architecture of AI services using Python, PostgreSQL, Redis Streams and microservices.

***Technologies: Python, LangChain, LangGraph, LLMs, NLP, PostgreSQL, Redis Streams, Docker, Microservices, Celery***

**Software Engineering Intern** **Jul 2025 – Sep 2025**

***National Advanced School of Engineering (ENSPY)*** Yaounde, Cameroon

- Developed a real-time communication platform supporting messaging and peer-to-peer video communication.
- Implemented real-time communication using Socket.IO and WebRTC protocols with sub-100ms latency.
- Contributed to backend architecture, REST APIs and real-time data synchronization.
- Collaborated in an Agile development environment through sprint planning, code reviews and technical discussions.

***Technologies: React.js, Node.js, Socket.IO, WebRTC, REST APIs, Flutter***

## SELECTED AI PROJECTS

**Autonomous Agentic ERP Bridge & Logistics Assistant** **HADRON S.A.**

- Event-driven autonomous system automating ERP decision-making (Inventory & Logistics) via intelligent reasoning agents.
- Engineered Redis Streams streaming pipeline, LangGraph contextual reasoning, and Pydantic deterministic safety guardrails.
- Maintains sub-300ms decision latency on 95% of operational events with a 99.2% automated validation rate.

***Technologies: Python, FastAPI, LangGraph, Redis Streams, PostgreSQL, Docker, Celery***

**Diabetes Prediction Using Machine Learning** Open Source / ML Lab

- Developed and compared multiple classification models including Logistic Regression, Support Vector Machines and Random Forest.
- Performed exploratory data analysis, feature analysis and preprocessing including missing-value handling, outlier analysis and feature scaling.
- Evaluated model performance using Accuracy, Precision, Recall, F1-score and ROC-AUC metrics.

***Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn***

## AI & MACHINE LEARNING KNOWLEDGE

**Machine Learning:** Supervised and Unsupervised Learning, Classification, Regression, Model Evaluation, Feature Engineering

**Deep Learning:** Neural Networks, CNNs, RNNs, LSTMs, Transformers, Spatial Attention

**Natural Language Processing:** Text Processing, Language Models, Large Language Models, Prompt Engineering, Retrieval-Augmented Generation

**Mathematical Foundation:** Linear Algebra, Calculus, Probability, Statistics and Optimization

## ADDITIONAL AI/ML TRAINING

**Self-Directed Study Using Stanford Online Resources** **2023 – Present**

- Machine Learning concepts inspired by CS229 (Supervised/Unsupervised Learning, Kernel Methods, Learning Theory).
- Deep Learning & Computer Vision concepts inspired by CS231n (Convolutional Networks, Spatial Features, Object Detection).
- Natural Language Processing concepts inspired by CS224n (Word Embeddings, Transformers, Attention Mechanisms, LLMs).

## LANGUAGES & CORE INTERESTS

French: Native English: Professional working proficiency

**Research Interests:** AI for Healthcare Diagnostics, Agricultural Optimization in Africa, Scalable ML Systems